Melody_002 v0.1 Validation Report

1. Test Identification

- **Melody**: melody_002
- **Source**: `app/tests/melodies/melody_002/Eb_major/audio.wav`
- **BPM used for quantization**: 120
- **Purpose**: v0.1 validation and regression documentation

2. Results that PASS

Raw Note Detection

- ✅ Completed successfully
- Detected 19 raw notes from audio
- First notes: D#4 (MIDI 63), G4, A#4, G#4, G4...

Enharmonic Pitch Class Handling

- ✅ Handled correctly at the key-detection level
- MIDI 63 notes appropriately labeled as D#4/Eb4 equivalents
- No unexpected pitch errors introduced

Key Detection

- ✅ Key detected as **Eb major**
- ✅ Mode: **major**
- ✅ Fifths detected as **-3** (correct for Eb major)

Regression Verification

- ✅ D major and E major sharp-key regression cases remain correct
- Verified via existing test suite: `app/tests/test_key_detection.py`

Quantization

- ✅ Completes successfully with BPM=120.0
- Produced 18 quantized notes after cleanup
- Sample quantizations:
- D#4: beat_start=0.02, beat_duration=0.98 → quarter
- G4: beat_start=1.00, beat_duration=1.00 → quarter
- A#4: beat_start=2.00, beat_duration=0.60 → eighth
- Final D#4: beat_start=14.01, beat_duration=4.23 → whole

Measure Validation (Partial)

- ✅ Measures 1–4 each contain exactly 4 beats
- Measure 1: 4.000 beats (6 notes)
- Measure 2: 4.000 beats (3 notes)

- Measure 3: 4.000 beats (6 notes)
- Measure 4: 4.000 beats (3 notes)

3. Accepted Behavior

Remaining Post-Cleanup Overlaps

The following three overlaps remain after monophonic cleanup: 1. A#4[2] ↔ G#4[3]: **46 ms** 2. C5[7] ↔ A#4[8]: **46 ms** 3. G4[14] ↔ F4[15]: **23 ms**

Acceptance Justification: These overlaps are below the existing 50 ms timing-jitter threshold used in the cleanup algorithm (see `debug_monophonic_cleanup.py` line 54: `if overlap_duration < 0.05:`). Therefore, they are correctly accepted as timing jitter rather than treated as cleanup failures for v0.1.

4. Known Failure

Measure Count Deviation

- **Expected**: Exactly 4 measures of 4/4 time signature
- **Actual**: 5 measures produced
- **Measure Details**:
- Measures 1–4: 4.000 beats each ✓
- Measure 5: 2.000 beats ❌

Root Cause Analysis

The fifth measure contains only a trailing D#4/Eb whole note: - Quantized as approximately **4.23-beat/whole-note** event - Begins around beat 14.01 - Extends to approximately beat 18.24 - Crosses the measure-4 boundary (which ends at beat 16.00) - Creates the extra 2-beat tail in measure 5 (beats 16.00-18.00)

Important Distinction: The `build_measures()` function is behaving consistently with its input and is **not currently identified as the root cause**. The upstream issue is the overlong final-note duration produced by the transcription/cleanup pipeline.

5. Scope Decision

No Additional Final-Note Heuristic

At this time, **no additional final-note heuristic is being added**. Specifically, we are **not implementing** rules such as: - `final duration = previous duration × 4` - Any other melody-specific duration cap

Reasoning: Such rules risk overfitting to melody_002 rather than representing a general v0.1 transcription principle. The current approach maintains algorithmic generality while documenting this known limitation for future investigation.

6. v0.1 Status Summary

✅ Enharmonic Key Spelling: VERIFIED AND SUCCESSFUL

- Key detection correctly identifies Eb major
- Fifths correctly reported as -3
- No regression in sharp-key cases (D major, E major)

⚠️ Final-Note Duration/Reverb Handling: KNOWN LIMITATION UNDER INVESTIGATION

- Overlong final note causes measure count deviation
- Understood as upstream transcription/cleanup issue
- Not attributable to measure builder
- Requires further study of note offset detection

🎯 Measure Builder: NOT CURRENTLY CONSIDERED THE ROOT CAUSE

- Functions correctly given its input
- Measure validation failure stems from input characteristics
- No measure builder modification indicated at this time

📊 Overall Validation Status

Do not claim the entire melody passes validation. Instead: - Enharmonic key spelling: **SUCCESSFUL** - Final-note duration handling: **KNOWN LIMITATION** - Measure builder: **NOT AT FAULT** - Regression protection: **MAINTAINED**

Report generated using existing observation harness: `debug_monophonic_cleanup.py` *No production code modified during validation Documentation created:* `melody_002_v0.1_validation.md`